TN SKILL PROJECT | IDEATION PHASE

# Auto Ticket Classification using Flow Designer

*Documenting Phase 1: Problem Definition, Context, and Architectural Bottlenecks*

**Phase:** 01 / Define Problem **Domain:** ITSM & Workflow Automation **Tool:** Flow Designer

## 1. PROJECT BACKGROUND & CONTEXT

In modern IT Service Management (ITSM) and Enterprise Resource Planning (ERP) environments, service desks process hundreds to thousands of service requests (tickets) daily. These incoming requests vary widely in complexity, ranging from simple password resets and access requests to critical database outages and network infrastructure failures.

Currently, traditional enterprise helpdesks rely heavily on **manual ticket triage**. When a user submits an incident report via email or portal, the ticket is initially funneled into an unassigned central queue. Dedicated Level 1 (L1) support personnel must manually inspect every ticket, analyze the user's description, decide on the appropriate category and urgency, and manually reassign the ticket to specialized technical groups.

> **CORE PROBLEM STATEMENT**
>
> The current manual ticket classification and routing mechanism creates operational friction, leading to severe queue latency, high error rates in ticket assignment, increased Mean Time to Resolve (MTTR), and frequent Service Level Agreement (SLA) breaches. Without automated decision workflows, IT operations suffer from inefficient resource utilization and escalating support costs.

## 2. KEY CHALLENGES IN CURRENT MANUAL OPERATIONS

- **High Manual Overhead:** L1 support agents spend significant working hours forwarding and classifying tickets rather than solving actual technical incidents.
- **Prolonged Ticket Latency:** Tickets sit idle in central unassigned queues during off-hours or peak spikes, delaying first response times.
- **Human Error & Incorrect Routing:** Ambiguous user descriptions lead to misinterpretation, causing tickets to be bounced ("ping-ponged") between teams.
- **Inconsistent Prioritization:** Subjective classification by different agents results in inconsistent urgency levels for identical technical issues.

## 3. OPERATIONAL IMPACT & BASELINE COMPARISON

To justify the transition to automated classification, the operational gaps of the current manual process must be measured against the target state powered by Flow Designer.

| METRIC / DIMENSION | MANUAL PROCESS (CURRENT STATE) | TARGET STATE (FLOW DESIGNER) |
|---|---|---|
| **First Response Time** | 30 mins to 4 hours (Queue dependent) | Instantaneous (< 10 seconds) |
| **Classification Accuracy** | ~60% – 70% (Subjective to L1 agent) | > 90% (Standardized logic rules) |
| **Routing Bounce Rate** | High (20-30% re-routing rate) | Minimal (< 5% misallocation) |
| **Operational Overhead** | High (Requires constant manual triage) | Low (Automated background logic) |
| **SLA Compliance Rate** | Frequent breaches during peak loads | Consistent high compliance |

## 4. ROOT CAUSE ANALYSIS (5-WHYS FRAMEWORK)

A 5-Whys analysis was conducted to pinpoint the foundational failure of the current helpdesk triage mechanism:

**1. Why is the overall Mean Time to Resolve (MTTR) high?**
*Because tickets experience extended delays before reaching the correct technical resolution team.*

**2. Why do tickets take long to reach specialist teams?**
*Because they sit in unassigned central holding queues waiting for manual review by L1 personnel.*

**3. Why is manual review required for every ticket?**
*Because the current service desk platform lacks automated rule engines to parse text upon creation.*

**4. Why are automated classification rules absent?**
*Because incoming requests are processed as raw unstructured data without automated workflow triggers.*

**5. Root Cause:**
*The absence of an integrated, low-code workflow engine (Flow Designer) capable of evaluating incident properties and executing instant routing actions.*

## 5. PROPOSED SOLUTION: FLOW DESIGNER AUTOMATION

The proposed project implements a low-code automated workflow using **Flow Designer**. The system will operate as an event-driven engine that activates immediately upon ticket creation.

### TRIGGER & PARSING

Detects new ticket creation, parses title, short description, user location, and service impact parameters using keyword and conditional logic rules.

### CATEGORIZATION LOGIC

Evaluates properties against pre-configured decision tables to accurately set Category, Subcategory, Urgency, and Priority fields.

### AUTOMATED ASSIGNMENT

Directly routes the ticket to specialized technical queues (e.g., Network Ops, Database, Security) without human intervention.

### NOTIFICATION & SLA

Triggers real-time alerts to the assigned support group and attaches appropriate SLA timers immediately upon dispatch.

## 6. PROJECT GOALS & SUCCESS METRICS

- **Eliminate Queue Delays:** Reduce initial ticket triage time from hours to under 10 seconds.
- **Enhance Categorization Precision:** Achieve at least 85–90% accurate first-time assignment to technical groups.
- **Mitigate SLA Violations:** Decrease response-related SLA breaches by over 40%.
- **Optimize Resource Allocation:** Reallocate L1 human resources from repetitive routing tasks to direct issue resolution.

## 7. PROJECT SCOPE DEFINITION

### IN-SCOPE

- Flow Designer workflow architecture and trigger definitions.
- Keyword parsing logic & decision rule tables.
- Automated group routing and notification actions.
- Testing using standard incident datasets.

### OUT-OF-SCOPE

- Physical hardware repair and field dispatch operations.
- Direct integration with non-standard legacy platforms without API access.
- Manual resolution of complex custom code bugs.